TWO BOOKS · BIOLOGY — companion lecture deck

Reading the corpus through a genome lens

The honest spine

The genome lens (letters ≈ bases, roots ≈ codons, words ≈ proteins) is a MEASUREMENT FRAME, not a design claim. Base composition is governed by common letters and richness by length — read both against size. The genuine signal is the GRAMMAR FOOTPRINT: di-codon bias and the $H_0 \rightarrow H_1$ conditional-entropy drop. Shuffle nulls throughout; no hidden code.

Two domains, one method

Every Qur'an number is computed live from Book6. Every genomics number is mainstream, shown in round form and labelled as reference — the analogy is a lens to think with, audited stage by stage, never evidence.

The Two Books — two revelations of one Author

عالم التدوين (the WORD) · عالم التكوين (the ACT)

The WORD — قول الله
The Qur'an: God's speech in language. The Book of SCRIPTURE.

The ACT — فعل الله
The Universe / life: God's deed. The Book of CREATION.

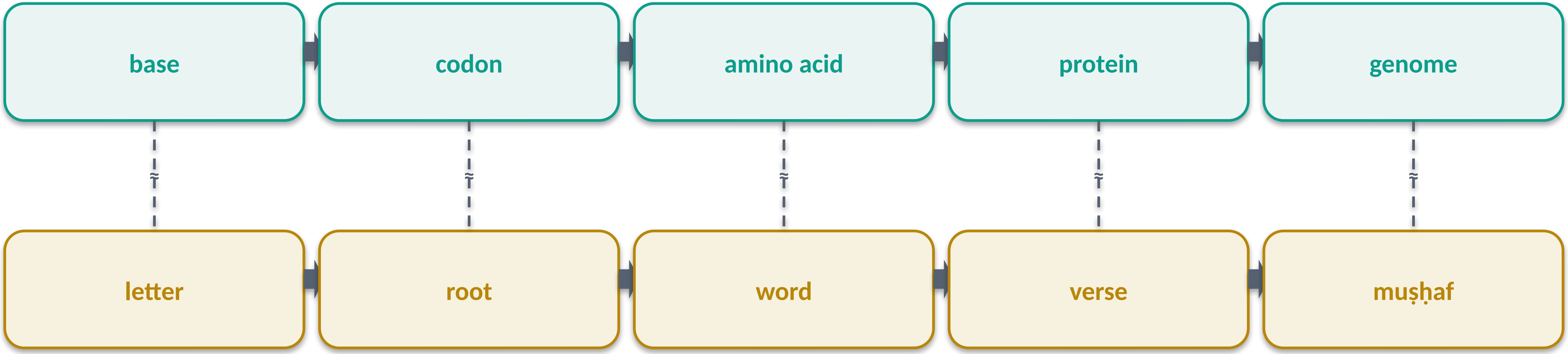
This deck borrows the genome's vocabulary to MEASURE the text

Genomics reads a sequence at nested levels — base, codon, amino acid, protein, genome. We map those levels onto the text's own — letter, root, word, verse, muṣḥaf — and import the matching tools: composition profiles, codon-usage curves, adjacency bias, conditional entropy.

We import the toolkit, not a claim that scripture is DNA. Every parallel is tested against a shuffle null and audited ✓ / ✗ / ~.

The unit ladder — genome ↔ text, level by level

GENOME (Creation) ≈ QUR'AN (Scripture) — matched levels



Matched levels, different substance

A clean ladder of correspondences in STRUCTURE — both build a vast lexicon from a tiny alphabet read in small groups. The deck tests each rung against a null; both ladders end in the human.

What this deck will — and will not — do

WILL

● Compute every Qur'an number live from Book6. ● Compare each to a shuffle null. ● Set it beside a mainstream genomics reference. ● Read composition AGAINST length, never raw. ● Audit ✓ / ✗ / ~.

WILL NOT

● Claim the text 'contains' DNA or a genetic code. ● Read base composition as a hidden cipher. ● Treat a length-driven richness as a choice. ● Offer the analogy as proof of anything.

The method — composition vs the length confound

The confound
Common letters and sūra length drive most of any composition number.

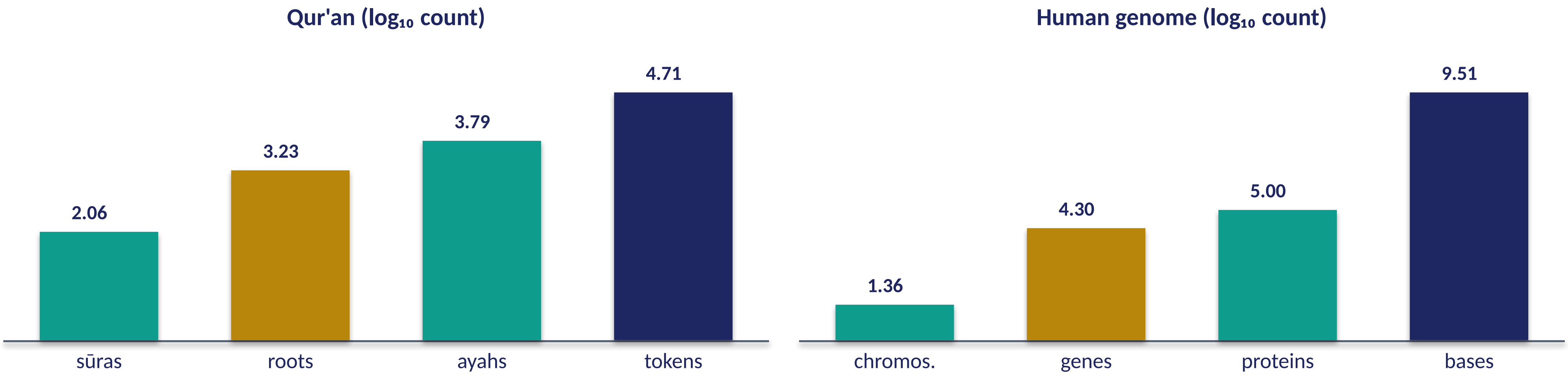
The fix
Read each value against the sūra's size; test the rest against a shuffle.

Two reflexes for every number

(1) NORMALIZE: a frequent letter or a low richness is usually just length and alphabet, not meaning. (2) SHUFFLE: scramble the sequence and rebuild the statistic; if the real value sits far in the tail, the ORDER carries structure the shuffle destroyed.

The genuine signals in this deck — di-codon bias and the $H_0 \rightarrow H_1$ entropy drop — are exactly the ones that survive the shuffle. Composition and richness are the ones we must read against size.

Two corpora by the numbers ($\log_{10}$)



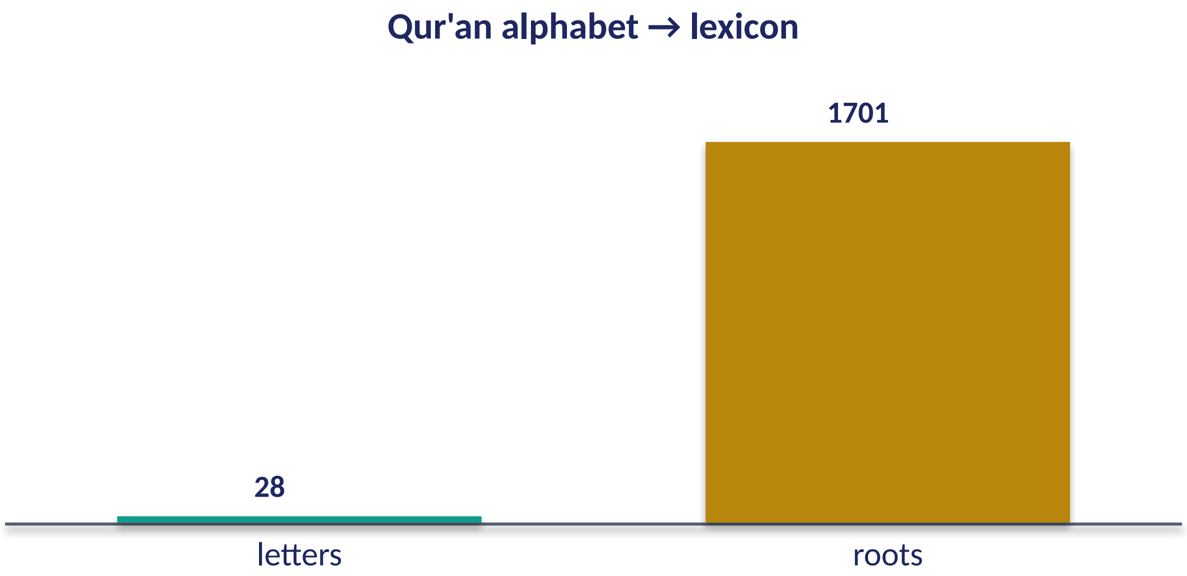
Qur'an — finite, countable

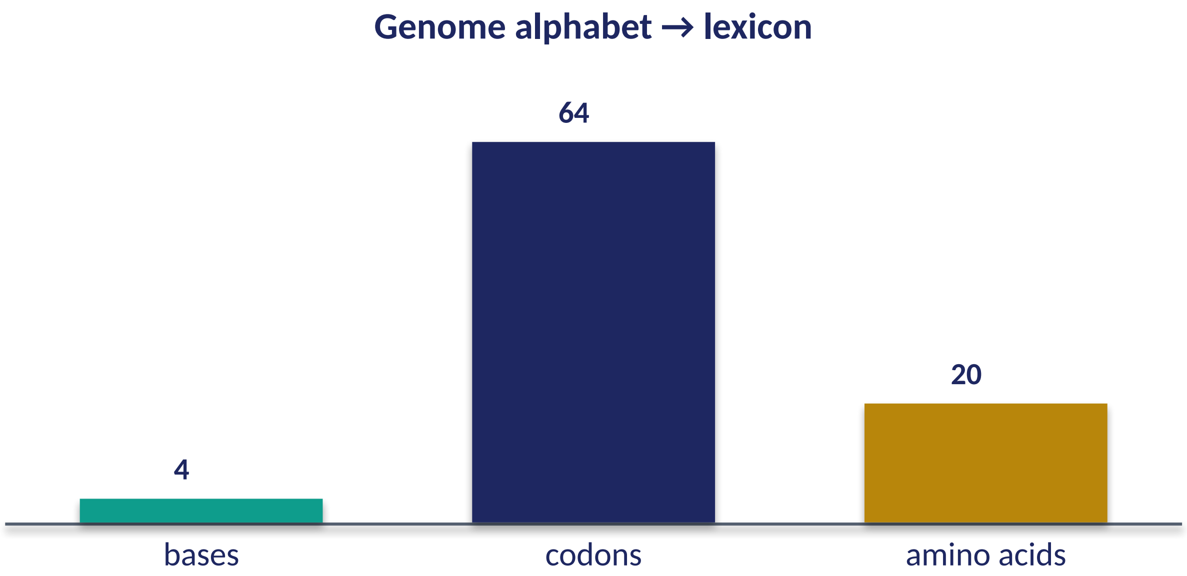
114 sūras · 6,236 ayahs · ~51k root-tokens · 1,701 roots. A corpus a person can hold in memory.

Genome — astronomically larger

23 pairs · ~20k genes · ~3.2 billion bases. The Book of Creation dwarfs the muṣḥaf (10^9 vs 10^4).

Module 1 — Frame: the genome lens





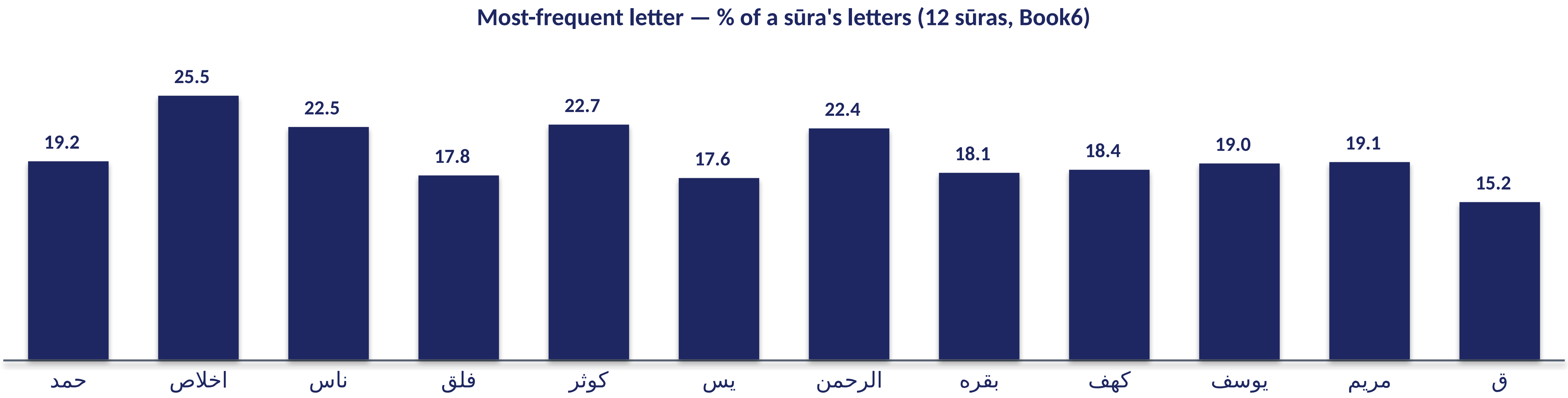
What it is / Why

Map letter→base, root→codon, word→protein, then apply genomics' counting + shuffle machinery. Both Books build a huge lexicon from a tiny alphabet read in small groups.

In the data · Bridge

~28 letters build 1,701 roots (most exactly 3 letters); 4 bases build 64 codons → 20 amino acids. Next: how often does each 'base' (letter) appear?

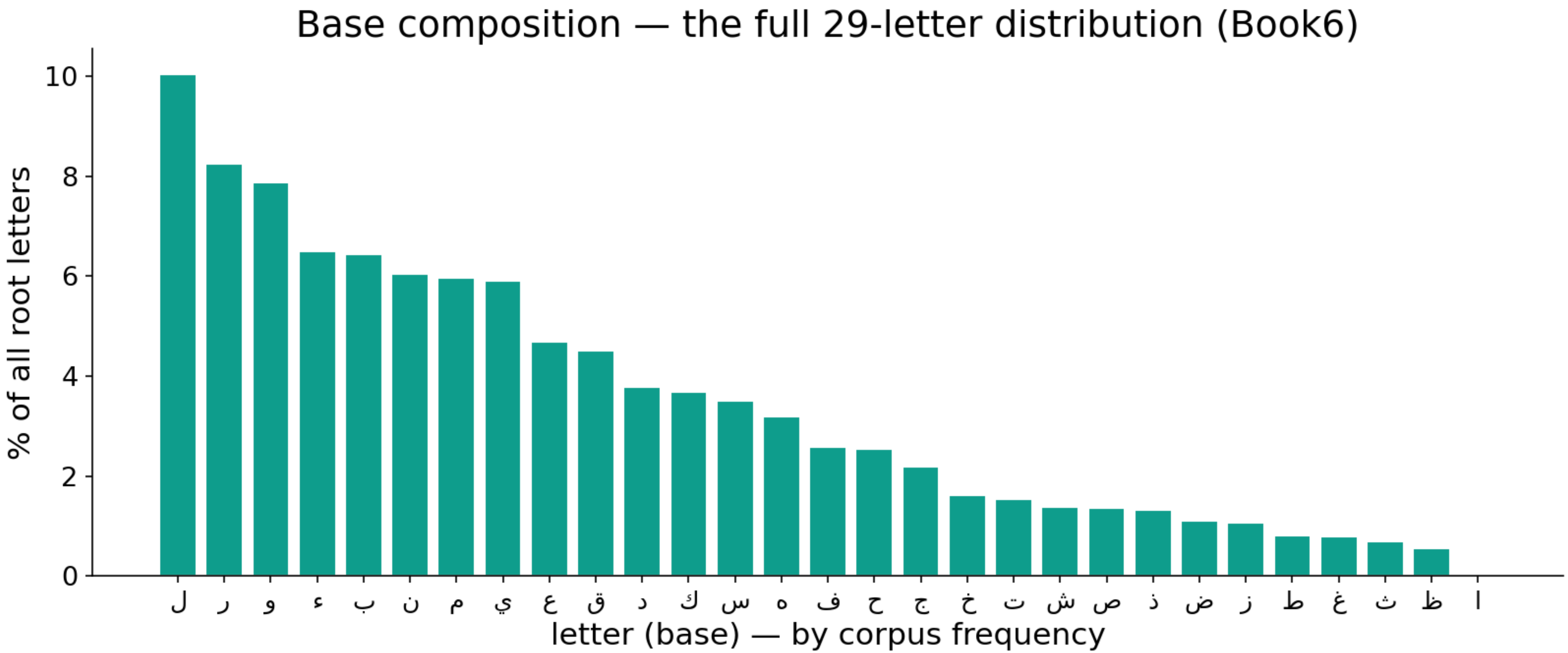
Module 2 — Base composition: the 'bases'



In the data — composition is dominated by common letters

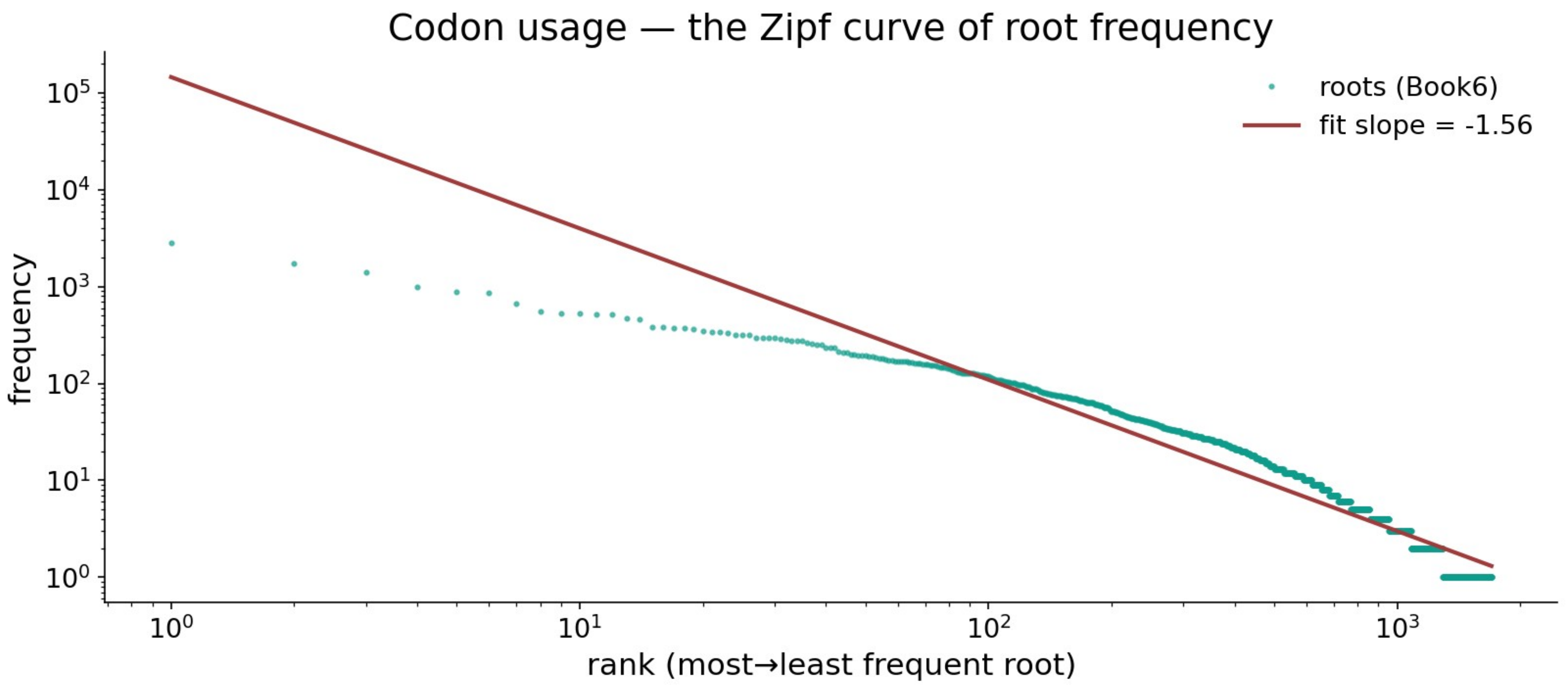
In al-Fatiha the single most frequent letter (ا) is 19.18% of its 146 letters. Across sūras the top-letter share is tightly clustered — every sūra draws on the same alphabet, so deviations from baseline are small. Read deviations, never raw counts; nothing here is a hidden code.

Module 2 — Base composition: the full letter distribution



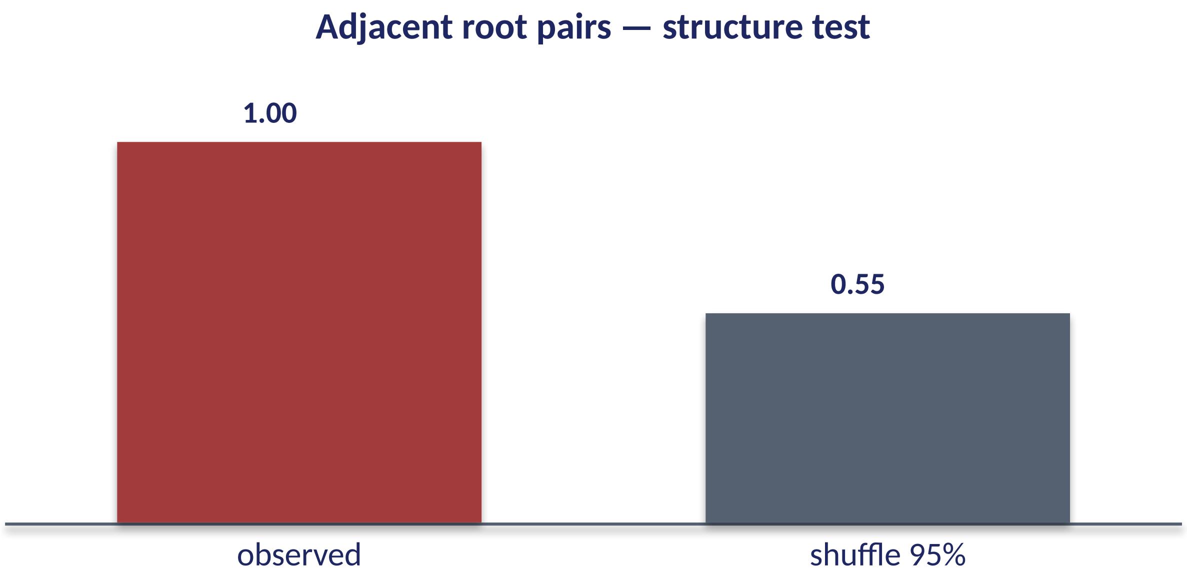
In the data — every letter's share of all root-letters (Book6). A few common letters dominate and the tail is long — composition is skewed by frequency, the baseline expectation, not a code.

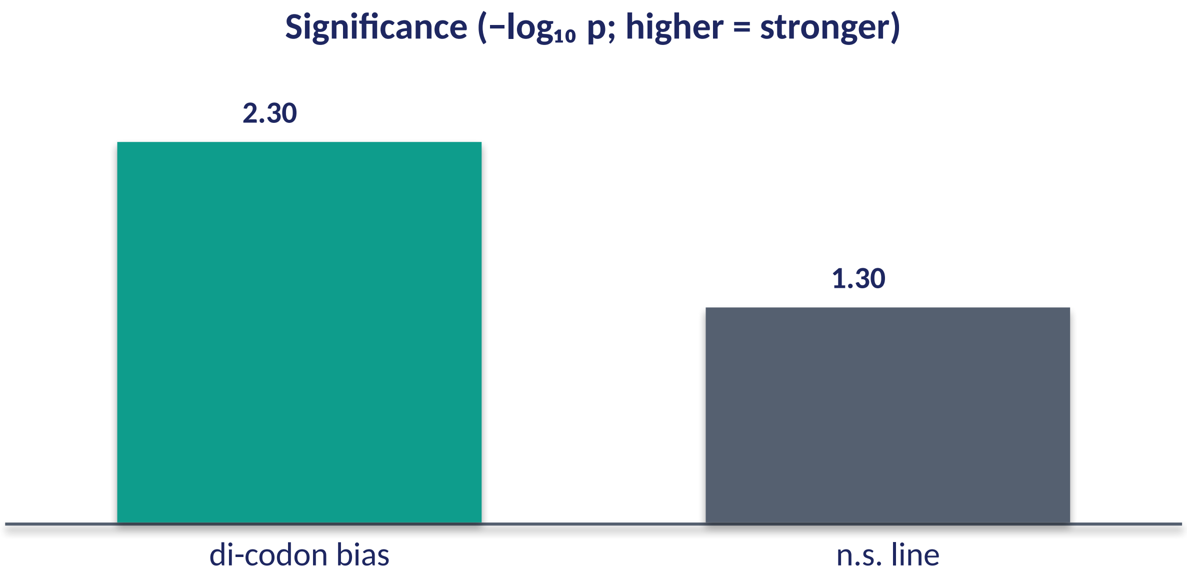
Module 3 — Codon usage: the Zipf curve



In the data — root frequency vs rank on log-log axes; the fitted slope ≈ −1.56 (steeper than ordinary word frequency ≈ −1) — a handful of roots dominate, a long tail barely appears.

Module 4 — Di-codon bias (adjacency structure)





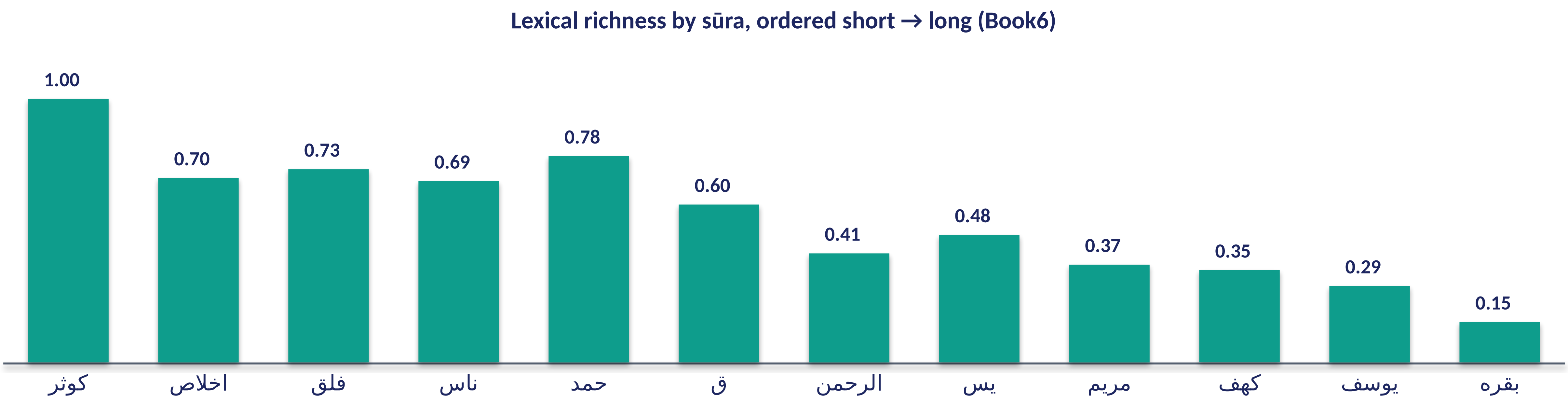
How / What we get

Tally observed adjacent root pairs, compare to a shuffle of the same roots, summarize the gap with a chi-square-like statistic and a permutation p.

In the data

Structure $p \approx 0.005$: adjacent pairs are significantly non-random — the footprint of grammar and fixed expressions. This is a GENUINE signal, distinct from mere composition.

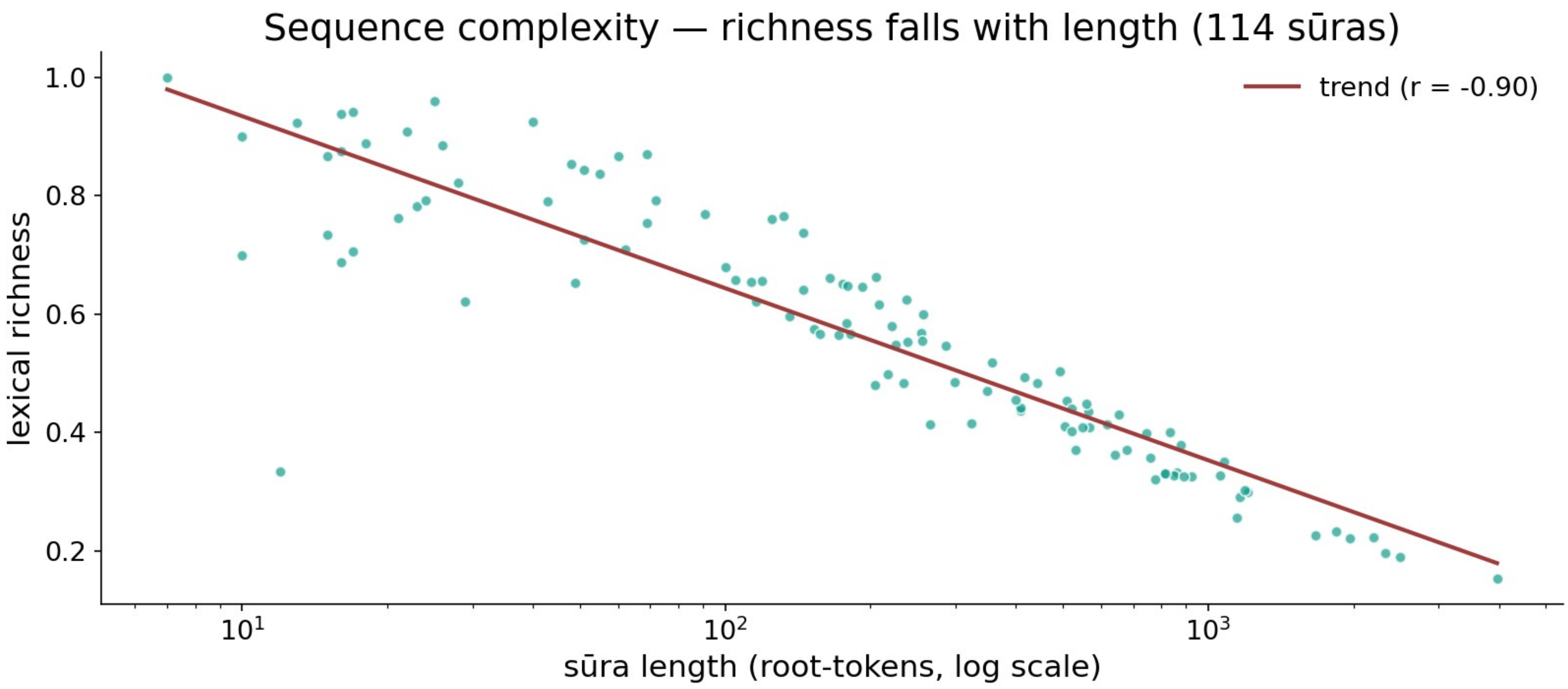
Module 5 — Sequence complexity: richness vs length



In the data — richness falls as length rises

Richness = distinct roots ÷ total root-tokens. al-Fatiha (short) scores 0.783; longer sūras repeat vocabulary and score lower. Ordered short→long, the bars trend DOWN — a length effect, not a stylistic choice. Always read richness against sūra size.

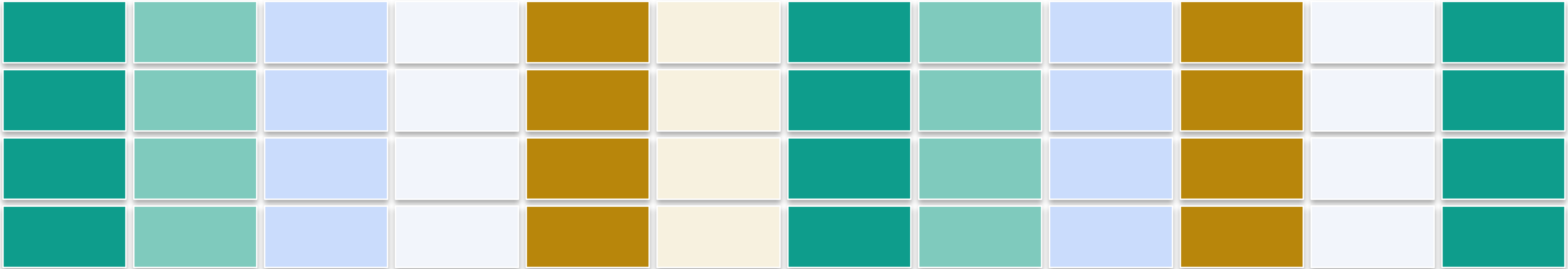
Module 5 — The length confound, in one scatter



In the data — every sūra's lexical richness vs its length (log scale), all 114 points; the trend is strongly negative (r ≈ −0.90) — richness is mostly a length artefact, not a stylistic choice.

Module 6 — Composition clustering (sūra as a vector)

each sūra → a vector of its top-root usage → grouped by similarity



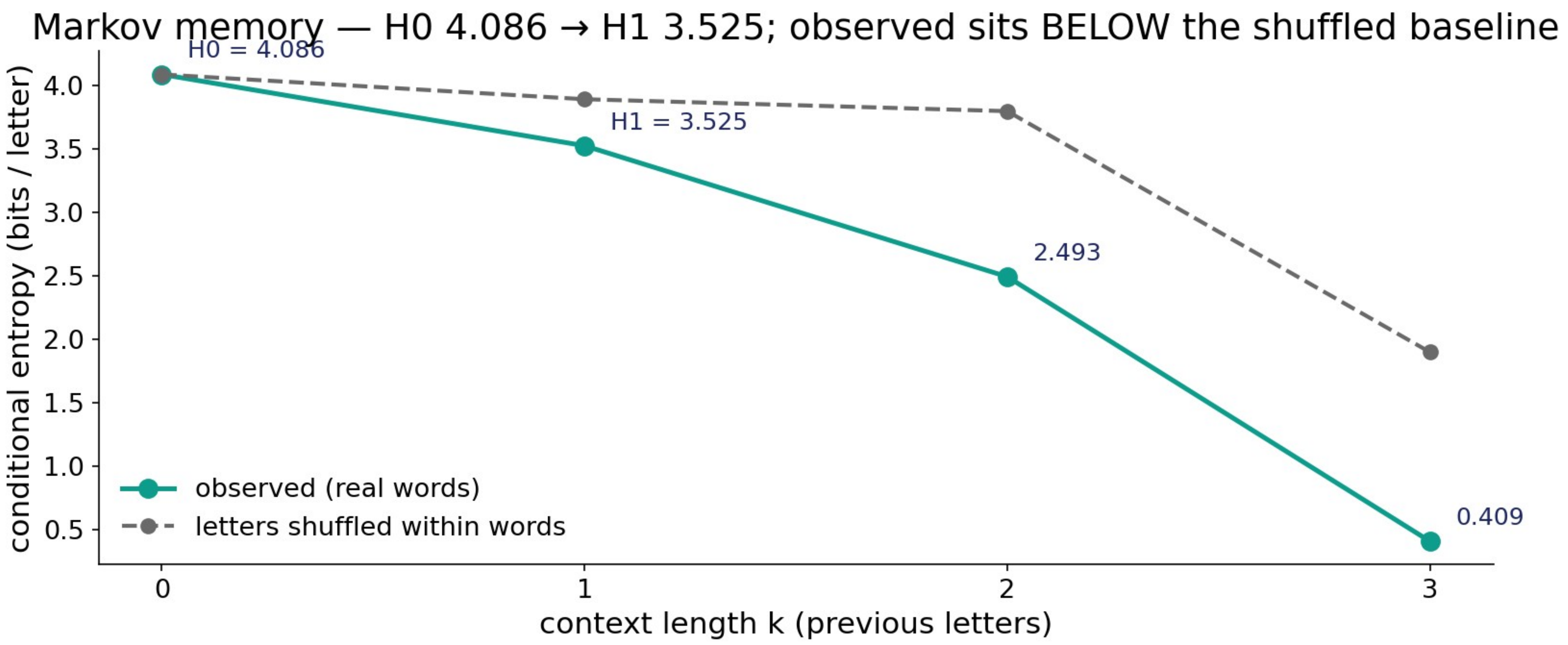
What it is / Why it matters

Represent each sūra as its usage of the top roots, then run Ward hierarchical clustering into a dendrogram. The branches group sūras with similar composition.

In the data — the clusters largely track STYLE and LENGTH: long Medinan sūras separate from short Meccan ones. Composition similarity is not evidence of hidden thematic coding.

Takeaway — sūras cluster by the mundane drivers of composition. Bridge: zoom back to the letter stream — how predictable is the next letter?

Module 7 — Markov memory: conditional-entropy decay



In the data — entropy of the next letter falls with context — H_0 4.086 → H_1 3.525 bits (≈ ½ bit gained). The observed curve sits BELOW the letters-shuffled baseline: real intra-word structure, the language footprint.

Module 8 — Synthesis: what the genome lens shows

Survives the null

Di-codon bias ($p \approx 0.005$) and the H_0 4.086 → H_1 3.525 entropy drop — a real GRAMMAR footprint, the same any natural language shows.

Driven by confounds

Base composition (common letters), Zipf skew (−1.56), and richness (falls with length) — read these AGAINST size, not as meaning.

Honest reading

The lens reveals ordinary language structure, not a hidden biological code. Say only what the data licenses.

Audit — the analogy, rung by rung

~ Base ↔ letter composition

Real but dominated by common letters.

✓ Codon usage ↔ Zipf

Heavy skew, slope ≈ −1.56.

✓ Di-codon ↔ grammar

Adjacency bias survives shuffle ($p \approx 0.005$).

~ Richness ↔ complexity

Mostly a length artefact; read vs size.

✓ Markov ↔ memory

$H_0 \to H_1$ drop is real language structure.

✗ Sequence ↔ genetic code

No codon table, no translation, no cipher.

Not a scientific miracle — and not evidence

What we are NOT claiming

The genome lens does not show that the Qur'an 'contains' DNA, encodes a genetic code, or was written biologically. letters are not bases, roots are not codons — those are LABELS on a measurement frame, not biological identities.

What the lens IS for

It is a disciplined way to MEASURE composition, adjacency, and memory, with a shuffle null behind every answer. It reveals an ordinary grammar footprint and the confounds of length and common letters — every number recomputable live in the app. The analogy is judged by clarity, never offered as proof.

Quick reference — terms & live numbers

Terms

Base composition — per-letter frequency. Codon usage / Zipf — root-frequency skew. Di-codon bias — adjacent-pair over/under-representation. Lexical richness — unique ÷ total roots. Conditional entropy — uncertainty of the next letter given the previous.

Live Book6 numbers

al-Fatiha top letter 19.18 = l% of 146 · Zipf slope −1.56 · di-codon $p \approx 0.005$ · richness 0.783 (falls with length) · H_0 4.086 → H_1 3.525 bits.

Close — one corpus, measured honestly

The through-line

Read as a genome, the Qur'an behaves like coherent natural language: a Zipfian lexicon over a small alphabet, composition set by common letters and length, real adjacency grammar (di-codon bias), and short-range memory ($H_0 \rightarrow H_1$) — but no genetic code and no hidden cipher.

Every claim carries a shuffle null and a live Book6 number; every cross-domain parallel is a labelled lens, audited ✓ / ✗ / ~, never evidence.

Next in the series — the FDR Summary that collects every Two Books test (Disjoint Letters · Signal · Biology) into one Benjamini–Hochberg-corrected dashboard, so no single p-value is read in isolation.